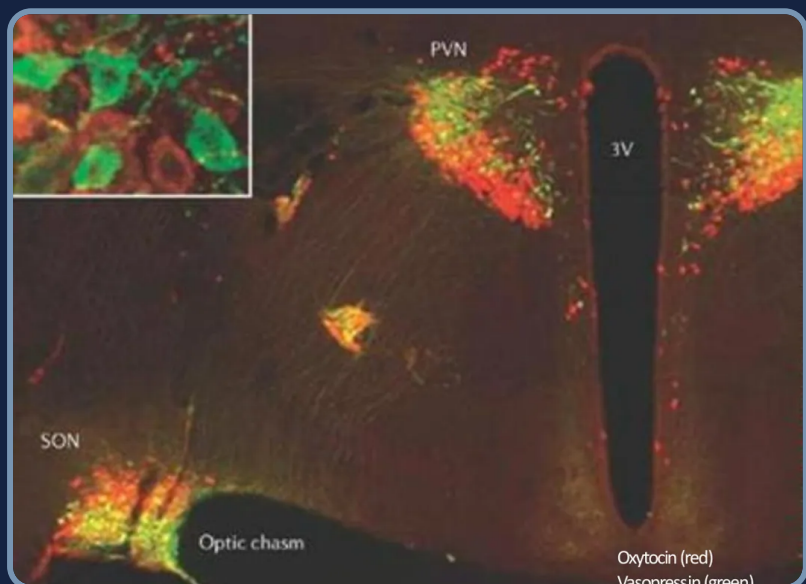


Hypothalamus & Pituitary Gland

The brain's hormone control room, explained with original lecture diagrams, memorable word origins, clinical contrasts and a detailed, compact exam sheet.

7 CHAPTERS / SLIDES 3-27 / RECALL PRACTICE



BEFORE YOU START

Your lecture, made memorable.

Use this guide alongside the 28-slide “Hypothalamus and Pituitary Gland” lecture. Every chapter is labelled with the original slide numbers. Course-specific figures are reproduced from the supplied lecture; etymology and memory stories are additional learning aids.

01 • Slides 3-5

The hypothalamus and its nuclei

02 • Slides 6-9

Two pituitary routes + hormone table

03 • Slides 10-13

Development, tissue and storage

04 • Slides 14-19

Vasopressin and water control

05 • Slides 20-22

DI and SIADH

06 • Slides 23-24

Receptors and vole behaviour

07 • Slides 25-27

Oxytocin, labour and milk ejection

EXAM DESK

Complete compact cheat sheet + recall

3 ways to remember

Decode the name: e.g. *aquaporin* = water pore. **Follow the mechanism:** AVP → V2 → cAMP → AQP2. **Explain the clinical opposite:** DI loses free water, SIADH retains it.

Teaching notes interpret the uploaded lecture; background word origins and analogies are supplemental memory aids. Diagnostic testing described here is not for self-administration.

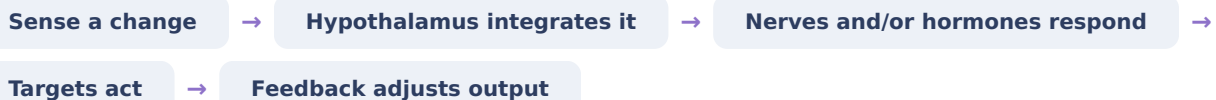
CHAPTER 01 · SLIDES 3-5

The brain's homeostasis control room

Hypothalamic location, inputs, nuclei & the meaning behind the names

Slide 3 · One tiny region, two giant jobs

The **hypothalamus** integrates signals from the central nervous system, peripheral nervous system and endocrine system. It coordinates **neural output through the autonomic nervous system** and **hormonal output through the pituitary**. The lecture links it to growth, fertility, body-water balance, temperature/metabolism, energy balance, stress and lactation.



✦ DECODE THE NAME: HYPO + THALAMUS

Hypo means “below”: the hypothalamus lies beneath the thalamus. Instead of memorising a floating brain label, picture a control room tucked *under* the thalamus, coordinating the body's internal conditions.

Slide 4 · Find the third ventricle

In the frontal (coronal) section, the hypothalamus is arranged on **both sides of the third ventricle (3V)**, beneath the thalamus. Its small size (the slide gives approximately 4 cm³ / 0.3% of brain mass) contrasts with its large range of functions.

Slide 5 · A nucleus is a team of neurons, not a cell nucleus

In the brain, a **nucleus** is a cluster of functionally related **neuronal cell bodies**. A single hypothalamic nucleus can contain different neuronal populations; the borders between nuclei may be indistinct. Some contain **neuroendocrine neurons** that communicate with the pituitary.

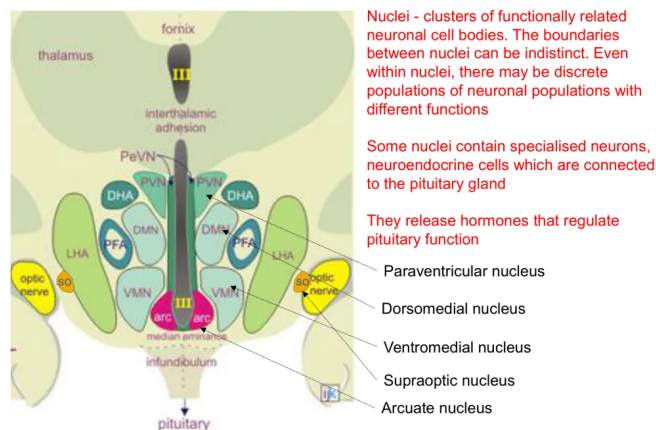
PVN · paraventricular nucleus

Para = **beside** the ventricle. The slide distinguishes PVN from the separate periventricular nucleus (PeVN).

SON · supraoptic nucleus

Supra = **above** the optic region. SON is one of the magnocellular hormone-producing areas in the lecture diagrams.

Coronal Section Through Hypothalamus



Nuclei - clusters of functionally related neuronal cell bodies. The boundaries between nuclei can be indistinct. Even within nuclei, there may be discrete populations of neuronal populations with different functions

Some nuclei contain specialised neurons, neuroendocrine cells which are connected to the pituitary gland

They release hormones that regulate pituitary function

✦ **LOCATION WORDS ARE BUILT-IN GPS**

Arcuate = curved like an arch; **ventromedial** = toward the belly/front and the midline; **dorsomedial** = toward the back and the midline. Decode labels into positions instead of learning them as unrelated syllables.

CHAPTER 02 · SLIDES 6-9

Two routes from brain to gland

Magnocellular vs parvocellular cells and the hormone traffic map

Slide 6 · Neuroendocrine means neuron + endocrine cell

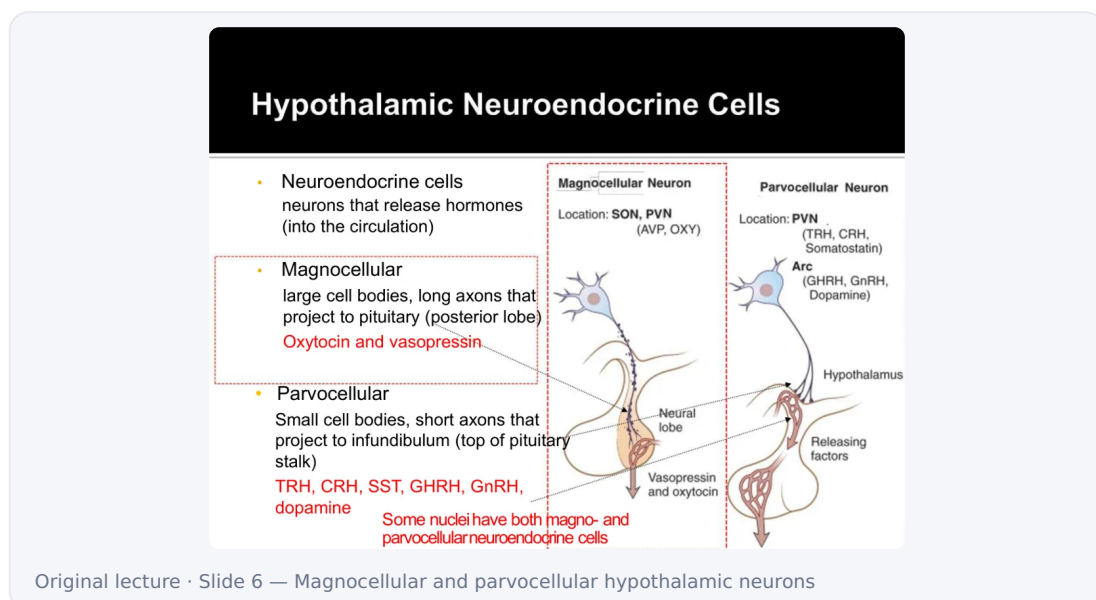
Magnocellular neurons have large cell bodies and long axons travelling to the **posterior pituitary**. They deliver oxytocin and vasopressin (AVP/ADH). **Parvocellular** neurons have smaller cell bodies and shorter axons terminating near the upper pituitary stalk; they release TRH, CRH, somatostatin, GHRH, GnRH and dopamine into the **hypophyseal portal circulation**. Some hypothalamic nuclei contain both types.

Magno = big / long

An axon is the delivery cable: hypothalamus → posterior pituitary → general circulation.

Parvo = small / short

The axon ends at the upper stalk: factors enter a local blood shortcut → anterior pituitary.



Slides 7-8 · Anatomy explains the different jobs

The **pituitary** (also called the **hypophysis**) sits below the hypothalamus. The **anterior lobe / adenohypophysis** is glandular and receives regulatory hormones via portal blood; the **posterior lobe / neurohypophysis** is neural and releases hormone delivered along axons from hypothalamic cell bodies. The optic chiasm is just above the gland.

Parvocellular neurons → Portal blood → Anterior pituitary cells → Pituitary hormone

Magnocellular neurons → Long axons → Posterior pituitary release → Systemic blood

✦ A NAME IS A MICROSCOPIC JOB DESCRIPTION

Adeno means gland; **neuro** means nerve. The anterior lobe contains endocrine *cells that make their own hormones*; the posterior lobe mainly contains nerve endings that *release hormones made in the hypothalamus*.

Slide 9 • The six controllers and the two posterior hormones

The six **hypophysiotropic** factors control **anterior** pituitary hormone release. The table below follows the slide's division: AVP and oxytocin instead belong to the **posterior** pituitary pathway.

Hypothalamic signal	Anterior pituitary effect	Name clue
TRH	↑ TSH	Thyro = thyroid
CRH	↑ ACTH	Cortico = adrenal cortex
Somatostatin (SST)	↓ GH	Statin = holding / stopping
GHRH	↑ GH	Growth-hormone releasing
GnRH	↑ FSH and LH	Gonado = gonads
Dopamine	↓ Prolactin	The prolactin brake

Hormones of Hypothalamus

Hypophysiotropic hormones

Hormone	Class	Function
Thyrotrophin Releasing Hormone (TRH)	Peptide (3 aa)	Stimulate TSH secretion from anterior pituitary
Corticotrophin Releasing Hormone (CRH)	Peptide (41 aa)	Stimulate ACTH secretion from anterior pituitary
Somatostatin (SST)	Peptide (14/28 aa)	Inhibits GH release from anterior pituitary
Growth Hormone releasing Hormone (GHRH)	Peptide (44 aa)	Stimulate GH secretion from anterior pituitary
Gonadotrophin Releasing Hormone (GnRH)	Peptide (10 aa)	Stimulate FSH and LH secretion from anterior pituitary
Dopamine	Catecholamine	Inhibits prolactin release from anterior pituitary
Vasopressin (AVP)	Peptide (9 aa)	Stimulates water reabsorption by kidneys
Oxytocin	Peptide (9 aa)	Stimulates uterine contractions during childbirth

Hormones of posterior pituitary

Original lecture · Slide 9 — Source table of hypothalamic hormones and their functions

✦ “TROPIC” IS ABOUT LOOKING AFTER ANOTHER TISSUE

Hypophysio-trophic signals regulate the hypophysis (pituitary). Here TRH, CRH, SST, GHRH, GnRH and dopamine act on the anterior lobe; AVP and oxytocin use the posterior route.

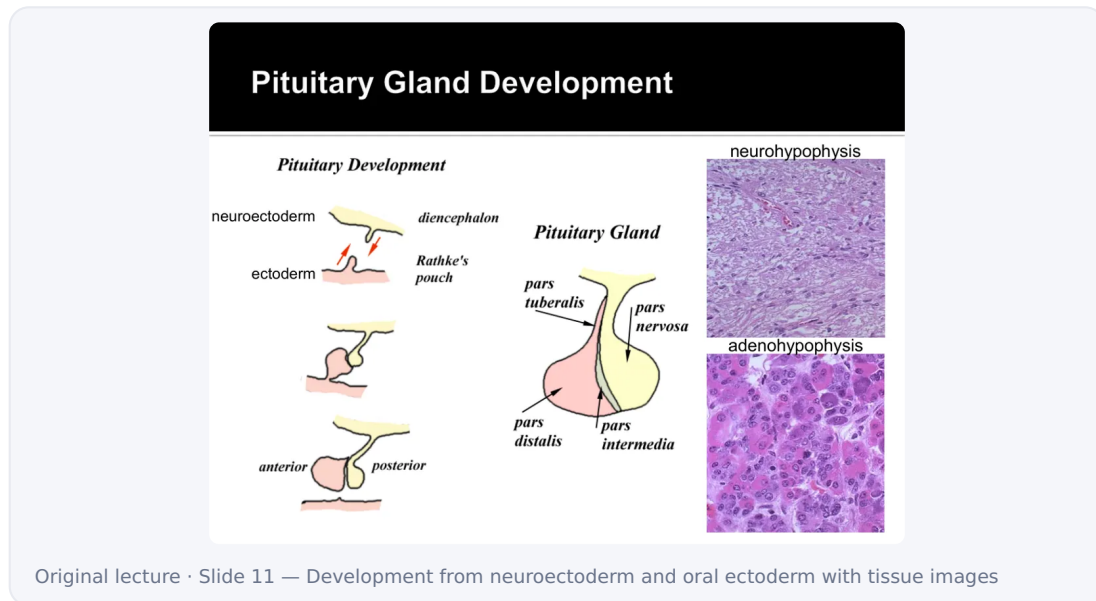
CHAPTER 03 · SLIDES 10-13

A pituitary built from two origins

Neural tissue, embryology, glia and microscopic hormone storage

Slides 10-11 · Why the back of the pituitary looks like brain tissue

Slide 10 starts the posterior pituitary section. Slide 11 shows the **neurohypophysis developing from neuroectoderm** extending down from the brain, while **Rathke's pouch** (oral ectoderm) develops into the **adenohypophysis**. The posterior lobe is therefore a neural continuation of the hypothalamus, not a second independent gland.



✦ RATHKE'S POUCH EXPLAINS TWO LECTURE TOPICS AT ONCE

The front comes from an upward-growing pouch near the developing mouth; the back grows downward from brain tissue. Remember **mouth meets brain** and you can reconstruct both lobes and understand why their histology differs.

Slide 12 · Posterior pituitary histology: fibres, helpers and packed parcels

Three visual clues identify posterior pituitary tissue: **unmyelinated axons** from hypothalamic neurons, supporting **pituicytes** (glial cells), and **Herring bodies**—local swellings in axons containing secretory granules with oxytocin or vasopressin.

Pituicytes = support cells

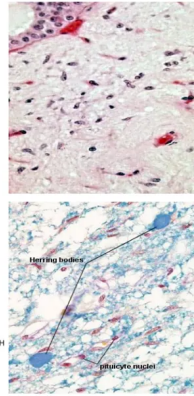
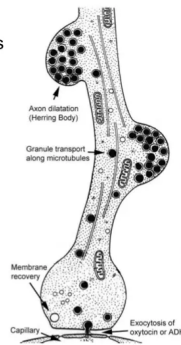
The **-cyte** ending means "cell"; these are glial cells of the posterior pituitary.

Herring bodies = storage stops

Picture a delivery van swelling with parcels: hormone-filled granules accumulate in axonal dilations before release near capillaries.

Functional Histology of Neurohypophysis

- Unmyelinated nerve axons of neurons cells whose cell bodies are located in hypothalamus
- Pituicytes (glial cells)
- Herring bodies
Secretory granules containing oxytocin and vasopressin in terminal region of axon



Original lecture · Slide 12 — Posterior pituitary histology: pituicytes, axons and Herring bodies

Slide 13 · Colour-code the neurons

The fluorescent image labels **oxytocin in red** and **vasopressin in green** in hypothalamic regions around the third ventricle (3V), PVN and SON. The image helps place the hormone-producing neurons in the brain rather than imagining they originate in the posterior pituitary.

CHAPTER 04 · SLIDES 14-19

Vasopressin: the water-saving switch

Receptors, osmosis, release triggers and aquaporin-2

Slide 14 · One molecule, two different receptors

Vasopressin (AVP) is also called **antidiuretic hormone (ADH)**. **V1 receptors** are abundant in vascular smooth muscle: receptor activation causes **contraction**. **V2 receptors** are abundant in kidney tubule epithelial cells: receptor activation promotes **water reabsorption**.

V1 · V for vessels

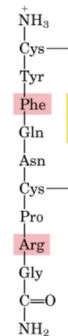
V1 → vascular smooth muscle contraction. “Vasopressin” links vessel action with its name.

V2 · H₂O rescue

V2 → renal water retention. “Anti-diuretic” = against making large amounts of urine.

Vasopressin

- Also known as antidiuretic hormone (ADH)
- Acts through 2 different receptors
 - V₁ receptors abundant in vascular smooth muscle cells
 - activation of V₁ receptors leads to VSMC contraction ('vasopressin')
 - V₂ receptors abundant in renal tubule epithelial cells
 - activation of V₂ receptors leads to increased renal H₂O reabsorption
- Plays essential role in maintaining water balance



Original lecture · Slide 14 — AVP structure and V1 versus V2 receptor actions

♦ WHY TWO NAMES FOR ONE HORMONE?

Vaso-pressin reflects blood-vessel constriction; **anti-diuretic hormone** describes the kidney effect that reduces urine volume. Same molecule, two target organs and two name clues.

Slides 15-16 · Why the brain cares about water

The lecture estimates that water makes up about **50-60% of body weight**. Body water determines **plasma osmolality** and therefore water movement across selectively permeable membranes. **Osmosis** is water moving across such a barrier; the diagrams show cells in isotonic, hypertonic and hypotonic surroundings.

Hypertonic outside

Water leaves the cell → cell shrinks.

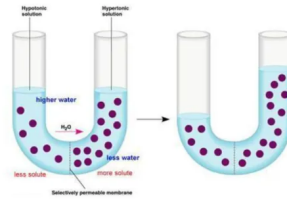
Hypotonic outside

Water enters the cell → cell swells. Isotonic: no net volume change.

Osmosis

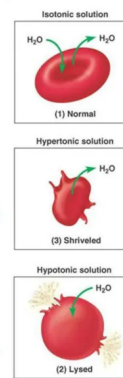
Movement of **solvent** molecules (H_2O) across a selective permeability barrier (eg. cell membrane, endothelium)

Body fluids (eg. blood, ECF) exert osmotic pressure (measured as osmolality)



The H_2O content of the body is the main determinant of osmolality
– regulation of body water content is known as **osmoregulation**

Osmolality is **tightly controlled to avoid this**



Original lecture · Slide 16 — U-tube osmosis demonstration and red blood cells in different tonicities

✦ OSMOSIS IN ONE PICTURE

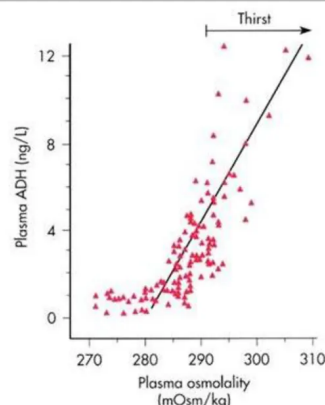
Imagine water trying to **dilute the saltier side** across a water-permeable barrier. Osmolality measures the concentration of dissolved particles; losing too much water makes body fluids more concentrated.

Slide 17 · A feedback loop, not just a hormone level

The lecture's graph shows AVP rising as plasma osmolality increases; it highlights a value of **>280 mOsm/kg** as a stimulus. AVP increases kidney water reabsorption, **decreasing urine volume and increasing urine concentration**; retained water helps lower plasma osmolality. Hypothalamic thirst centres also promote drinking.



Plasma AVP and Osmolality



Plasma osmolality of **> 280 mOsm/kg** induces AVP release

AVP acts on renal epithelial cells to promote reabsorption of H_2O

Plasma volume is increased and its osmolality decreased and urine volume is decreased and its osmolality is increased

Independently, thirst centres in the hypothalamus are activated increasing the urge to consume H_2O

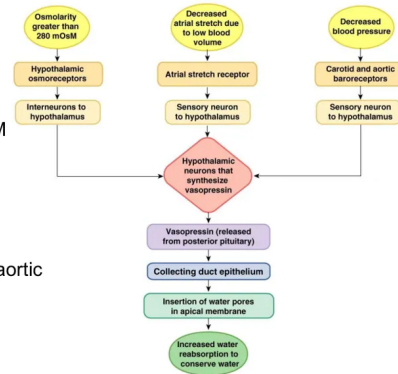
Original lecture · Slide 17 — Plasma osmolality plotted against AVP, with thirst response

Slide 18 · Three roads to AVP release

The lecture lists **higher plasma osmolality** sensed by hypothalamic osmoreceptors, **hypovolemia** sensed through lower atrial stretch, and **hypotension** sensed through carotid/aortic baroreceptors. Different sensors converge on hypothalamic AVP-producing neurons.

Factors Affecting AVP Release

- 3 stimuli control AVP release
- blood osmolality > 280mOsm (sensed by hypothalamus osmoreceptors)
- hypovolemia (< atrial stretch receptor activation)
- hypotension (< carotid and aortic baroreceptor activation)



Original lecture · Slide 18 — Flowchart of osmoreceptor, atrial stretch and baroreceptor inputs

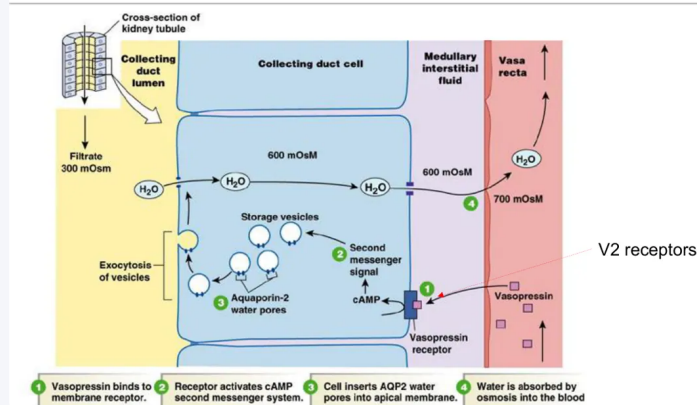
Slide 19 · How V2 makes the kidney a water-saving filter

Follow the numbered collecting-duct diagram: AVP binds **V2** on the epithelial cell → a **cAMP** second-messenger signal activates → vesicles insert **aquaporin-2 (AQP2)** water pores into the apical membrane → water enters the cell from tubular fluid and returns toward blood down the osmotic gradient.

AVP → V2 → cAMP signal → AQP2 inserted → H₂O crosses cell →

H₂O returns to blood

AVP Actions in Kidney Tubule Epithelia



Original lecture · Slide 19 — Numbered collecting duct mechanism showing cAMP and aquaporin-2 insertion

✦ AQUAPORIN IS THE WHOLE MECHANISM IN ONE WORD

Aqua = water; **porin** = pore. ADH does not simply shout “save water”: it helps insert actual water channels into a collecting-duct membrane so the water can move.

CHAPTER 05 · SLIDES 20-22

When water balance goes off-script

Diabetes insipidus, water-deprivation testing and SIADH

Slide 20 · Diabetes insipidus: too much dilute urine

In **diabetes insipidus (DI)**, the kidney fails to conserve enough water, producing large volumes of dilute urine (the slide gives an extreme example of **25 L/day**). **Central / cranial DI** involves inadequate AVP release due to a hypothalamic/pituitary problem; **nephrogenic DI** means the kidney does not respond adequately to AVP.

Central DI · missing message

Not enough usable AVP arrives from the hypothalamus/posterior pituitary route.

Nephrogenic DI · ignored message

AVP is present but renal response is deficient; the lecture mentions V2 receptor / AQP2 mutations and some drug causes.

Diabetes Insipidus (DI) : AVP Deficiency or Resistance

- Pathophysiology
 - inability of kidney to reabsorb H₂O with excretion of large volumes of dilute urine (25l/day)
- Types
 - neural (cranial) DI – hypothalamus/pituitary lesion
 - nephrogenic DI – renal 'resistance' to AVP
 - inherited - mutations in V2 receptor (X linked) or aquaporin 2 (autosomal)
 - acquired - drug induced (amphotericin B)
- Diagnosis
 - water deprivation test
- Treatment
 - desmopressin (synthetic AVP)

Original lecture · Slide 20 — Lecture summary of central and nephrogenic diabetes insipidus

✦ WHY "INSIPIDUS"?

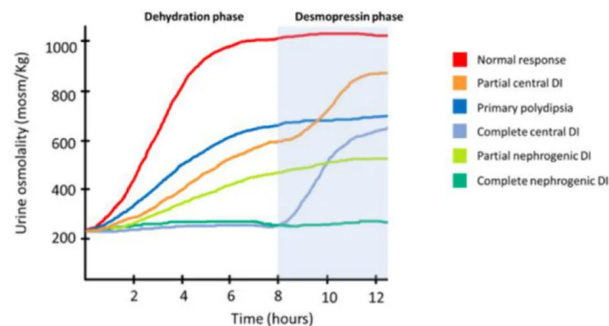
Historically **diabetes** referred to excessive urination. **Insipidus** means "tasteless", contrasting with **mellitus** ("honey-sweet"). They share the symptom of excess urine, not the same basic cause.

Slide 21 · Water deprivation asks two separate questions

The slide's **supervised** water-deprivation graph compares urine osmolality during dehydration, then after **desmopressin** (an AVP analogue). Normally urine becomes concentrated during dehydration. In central DI, desmopressin can increase urine osmolality because it supplies the missing AVP signal. In nephrogenic DI, the response remains limited because kidney responsiveness is impaired. Partial forms show intermediate responses.

Water Deprivation Test

Deprive subjects of water for ~ 12 hrs and measure urine osmolality/volume
Administer desmopressin at 8hrs



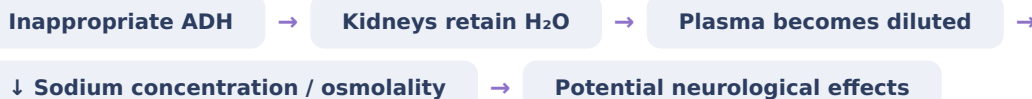
Original lecture · Slide 21 — Water deprivation and desmopressin response curves for DI subtypes

Clinical safety

The water-deprivation test is a medically supervised diagnostic procedure; the lecture graph is an explanation, not an instruction to perform a test yourself.

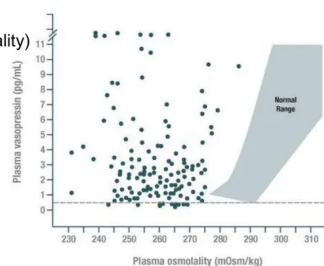
Slide 22 · SIADH: water is retained when it should not be

SIADH is inappropriate antidiuretic hormone secretion despite normal or increased plasma volume. Continued renal water reabsorption lowers plasma osmolality and can cause **hyponatremia** (low sodium concentration). The lecture warns that marked reductions in sodium/osmolality can produce brain swelling. It mentions ectopic ADH production by some tumours and a water-loading test as related clinical concepts.



Syndrome of Inappropriate Antidiuretic Hormone Secretion (SIADH)

- Inappropriate ADH secretion despite normal or increased plasma volume
 - ↑ H₂O reabsorption (↓ plasma osmolality)
 - normo or hypervolemia (hypertension)
 - hyponatremia
 - low sodium and plasma osmolality can lead to acute edema of the brain
- Etiology
 - ectopic production of ADH by GIT and pulmonary tumours
- Water loading test
 - tests the kidney's ability to excrete a water load by suppressing ADH



Original lecture · Slide 22 — SIADH findings and the plot of AVP versus plasma osmolality

✦ DI VS SIADH = OPPOSITE SIDES OF THE TAP

DI: too little effective antidiuretic action → **water escapes** as dilute urine. **SIADH:** too much inappropriate antidiuretic action → **water stays** and dilutes plasma sodium.

CHAPTER 06 · SLIDES 23-24

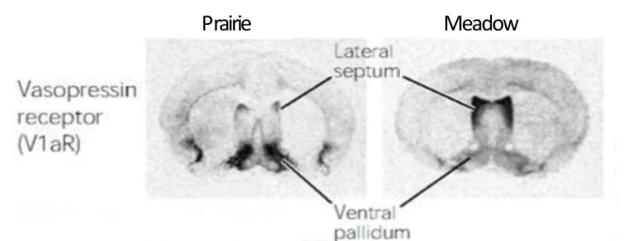
The vole story: receptors and behaviour

What an animal experiment can—and cannot—tell us

Slide 23 · Same hormone, different receptor map

The lecture compares receptor expression in the brains of **prairie voles** and **meadow voles**. It highlights the **ventral pallidum**, a brain region associated with reward, motivation and bonding. The slide links species differences in vasopressin-receptor patterns to differing partner-preference behaviour.

Patterns of AVP Receptor Expression in Brains of Meadow and Prairie Vole



Ventral pallidum is an area associated with trust, reward, motivation and bonding

Nair H P, Young L J Physiology 2006;21:146-152

Meadow vole = promiscuous
Prairie vole = faithful

Original lecture · Slide 23 — Prairie and meadow vole brain sections with different V1a receptor patterns

Slide 24 · Change one receptor map and observe a behaviour change

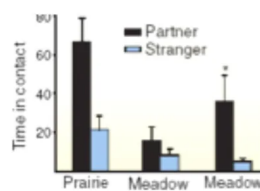
The lecture cites a **2004 Nature** study in which increasing vasopressin V1a-receptor expression in the ventral pallidum of male meadow voles was associated with increased **partner preference** compared with controls. The experiment illustrates that **where a hormone's receptor is expressed matters**, not just how much hormone is in circulation.

Vasopressin – A Cure for Infidelity ?

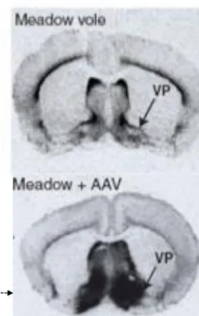
Enhanced partner preference in a promiscuous species by manipulating the expression of a single gene Nature 429, 754-757 (17 June 2004)

By transferring the vasopressin receptor gene to the pleasure centre of the naturally promiscuous male meadow vole, researchers were able to make it happily monogamous, a pattern of behaviour typical of its close relation, the prairie vole

Increased similarity to prairie vole behaviour



Increased AVP receptor expression in VP due to viral mediated gene transfer



Original lecture · Slide 24 — Meadow vole receptor-expression experiment and partner-preference graph

✦ THE INTERESTING POINT IS RECEPTOR GEOGRAPHY—NOT A “FAITHFULNESS DRUG”

A receptor changes what a tissue can *hear*. An animal brain circuit with more receptors may respond differently to the same chemical messenger. This vole experiment does **not** establish a treatment for human relationships or prove that one molecule dictates complex human behaviour.

CHAPTER 07 · SLIDES 25-27

Oxytocin: squeeze, release and repeat

Labour, milk let-down and the limits of the “cuddle chemical” label

Slide 25 · A close chemical cousin of AVP

Oxytocin is structurally similar to AVP; both are **9-amino-acid peptide hormones** in the lecture. Oxytocin's main physiological roles here are **parturition (childbirth)**, **lactation** and some social/mating/emotional behaviour. The slide notes weak antidiuretic action related to its structural similarity to AVP.

Oxytocin

- Structurally related to AVP
- Weak antidiuretic action (similarity to AVP)
- Physiological roles
 - parturition
 - lactation
 - social / mating / emotional behaviour

Original lecture · Slide 25 — Oxytocin peptide structure and physiological roles

✦ THE NAME GIVES AWAY THE JOB

Oxy comes from Greek for “quick/sharp” and **tokos** relates to childbirth. The name is a reminder of its role in strong uterine contractions; **parturition** simply means giving birth.

Slide 26 · Milk production ≠ milk ejection

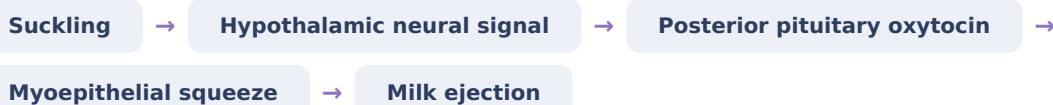
During suckling, nipple mechanoreceptors send signals to the hypothalamus, triggering **posterior pituitary oxytocin release**. Oxytocin contracts the **myoepithelial cells** around breast alveoli, pushing milk into the ducts: the **milk let-down reflex**. Oxytocin also stimulates uterine contractions during labour and may be administered clinically to induce labour.

Prolactin (anterior)

Makes the milk. Dopamine inhibits prolactin release; suckling can remove that brake.

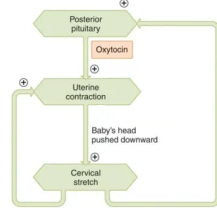
Oxytocin (posterior)

Moves the milk. It contracts myoepithelial “squeeze cells” around alveoli.

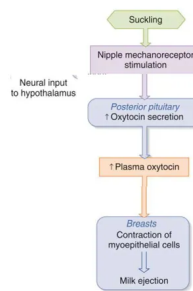


Physiological Role of Oxytocin

- Lactation
 - stimulates myoepithelial contractions to expel milk from mammary alveoli into ducts/collecting chamber ('milk let down reflex')



- Parturition
 - stimulates uterine contractions
 - can be used to induce labour



Original lecture · Slide 26 — Milk let-down and positive feedback in parturition

✦ A TINY CELL NAME IS A WHOLE CARTOON

Myo- = muscle; **epithelial** = part of a lining. Myoepithelial cells wrap around milk-filled alveoli like little hands squeezing a balloon. **Prolactin fills; oxytocin spills.**

Slide 26 · Childbirth as positive feedback

The lecture's diagram shows the fetus pressing against the cervix → **cervical stretch** stimulates oxytocin release → stronger uterine contractions push the baby downward → **more cervical stretch**. This is a **positive-feedback loop** that intensifies during labour.

Slide 27 · “Cuddle chemical” is a nickname, not a complete mechanism

The slide describes research-interest links between oxytocin and bonding, trust, anxiety and social behaviour. Those effects are context-dependent and more complex than a universal “trust hormone” response. For this lecture, prioritise the strongly established physiological mechanisms: **uterine contraction and milk ejection**.

The complete exam sheet

All the core lecture relationships in eight compact blocks · slides 3–27

01 · Map the brain · 3–5

Hypothalamus = neural + endocrine interface; integrates CNS/PNS/endocrine inputs; ANS output plus pituitary-mediated endocrine output. On either side of **3rd ventricle** beneath thalamus. **Nucleus** = cluster of functionally related neuronal bodies; PVN ≠ PeVN; SON supraoptic; ARC arcuate. Functions: growth, reproduction, osmoregulation, temperature/metabolism, energy, stress, lactation.

02 · Two secretory routes · 6–8

Magno: large cell body / long axon → posterior pituitary → systemic blood → AVP + oxytocin. **Parvo:** smaller / short axon → median eminence / upper stalk → portal veins → anterior pituitary regulation. **Adenohypophysis** = glandular anterior; **neurohypophysis** = neural posterior. Posterior releases hormones *made by hypothalamic neurons*.

03 · All 8 hypothalamic signals · 9

Anterior: TRH → ↑ TSH; CRH → ↑ ACTH; GHRH → ↑ GH; SST → ↓ GH; GnRH → ↑ FSH/LH; dopamine → ↓ prolactin.

Posterior: AVP/ADH → kidney H₂O conservation (also V1 vascular effect); oxytocin → uterine contraction and milk ejection. Slide 9 gives peptide lengths: TRH 3 aa, CRH 41, SST 14/28, GHRH 44, GnRH 10, AVP and oxytocin 9; dopamine is a catecholamine.

04 · Embryology / histology · 10-13

Oral ectoderm / Rathke's pouch → **anterior**; **neuroectoderm / infundibulum** → **posterior**. Posterior histology: unmyelinated axons from hypothalamus, glial **pituicytes**, **Herring bodies** = axonal secretory-granule accumulations. Fluorescence slide 13: oxytocin red, vasopressin green in PVN/SON.

05 · AVP and water homeostasis · 14-18

V1 vascular smooth-muscle contraction; **V2** kidney epithelial H₂O reabsorption. H₂O ~50–60% body mass. Osmosis = H₂O movement across selective membrane; hypertonic outside → cell shrink, hypotonic → swell. Lecture graph: ↑ osmolality (>280 mOsm/kg cited) → ↑ AVP and thirst; other stimuli: ↓ atrial stretch (hypovolemia), ↓ carotid/aortic baroreceptor firing (hypotension). AVP → ↓ urine volume / ↑ urine osmolality; reabsorbed H₂O → ↓ plasma osmolality.

06 · The collecting-duct mechanism · 19

AVP → basolateral **V2 receptor** → **cAMP** signal → **AQP2** vesicles insert channels into apical membrane → H₂O enters collecting-duct cells down osmotic gradient → H₂O returns to blood. AQP2 = aquaporin-2 water pore; the name literally describes the mechanism.

07 · Too little vs too much action · 20-22

Central DI: inadequate AVP output (hypothalamus/pituitary) → lots of dilute urine; desmopressin generally raises urine concentration. **Nephrogenic DI:** renal resistance (V2/AQP2 mutations; some acquired causes) → limited desmopressin response. **Water-deprivation test:** supervised assessment of urine concentration before/after desmopressin. **SIADH:** inappropriate ADH → water retention → ↓ plasma osmolality, hyponatremia, risk of cerebral edema; tumour-related ectopic ADH is one cause in slide.

08 · Receptors and reproductive physiology · 23-27

Prairie/meadow vole comparisons: different V1a receptor brain distribution; experimental V1a expression in meadow vole ventral pallidum ↑ partner preference (animal research, not proof of human “faithfulness”). **Oxytocin:** 9-aa peptide related to AVP. **Suckling** → **posterior oxytocin** → **myoepithelial contraction** → **milk ejection**. Contrast **anterior prolactin** → **milk synthesis**. Labour: cervical stretch → oxytocin → uterine contractions → more stretch (positive feedback). Social effects are complex/context-dependent.

Last-page self-test

Say each answer before reading the line under it. Then find the mechanism in the numbered lecture slide.

1. Why are there two pituitary routes?

Posterior receives hypothalamic long axons; anterior receives parvocellular signals through portal blood.

2. What is the difference between a brain nucleus and a cell nucleus?

A brain nucleus is a cluster of neuronal cell bodies.

3. Name all six anterior-pituitary regulatory factors.

TRH, CRH, SST, GHRH, GnRH and dopamine.

4. How can posterior tissue contain hormones it does not synthesise?

Hypothalamic neurons synthesise AVP/oxytocin and deliver granules along axons; Herring bodies store them.

5. Why can AVP be called both vasopressin and ADH?

V1 causes vascular smooth muscle contraction; V2 conserves water in the kidney.

6. How does AVP make water cross the collecting duct membrane?

V2 → cAMP → apical aquaporin-2 insertion → water reabsorption.

7. How does central DI differ from nephrogenic DI?

Missing/insufficient AVP signal versus kidney resistance to AVP.

8. Why does SIADH cause low plasma sodium?

Retained water dilutes plasma sodium and reduces osmolality.

9. What did the vole experiment show?

Increasing V1a receptor expression in a reward-related brain region increased partner preference in meadow voles, not proof of a human treatment.

10. During breastfeeding, which hormone makes milk and which ejects it?

Anterior prolactin synthesises milk; posterior oxytocin contracts myoepithelial cells for let-down.

11. What makes childbirth a positive-feedback loop?

Stretch → oxytocin → contraction → more cervical stretch.

You can find the complete 20-card deck and 16-question multiple-choice quiz in the accompanying website.